

Example 1

```
supplement_heart_rate |> anovas(outcome = resting_heart_rate_bpm,  
                                A = treatment,  
                                B = pony_type,  
                                interaction = TRUE)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	1	215.2813	215.2813	15.409	< 0.001
pony_type	2	1383.6803	691.8401	49.519	< 0.001
treatment:pony_type	2	636.6508	318.3254	22.785	< 0.001
Residuals	66	922.0942	13.9711		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(2, 66) = 22.785$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
supplement_heart_rate |> posthoc(outcome = resting_heart_rate_bpm,  
                                 A = treatment,  
                                 B = pony_type,  
                                 interaction = TRUE)
```

Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA ($\text{resting_heart_rate_bpm} \sim \text{treatment} * \text{pony_type}$)

Main effect: treatment

Comparison	Difference	p-value
Supplement vs Placebo	-3.4583	< 0.001 *

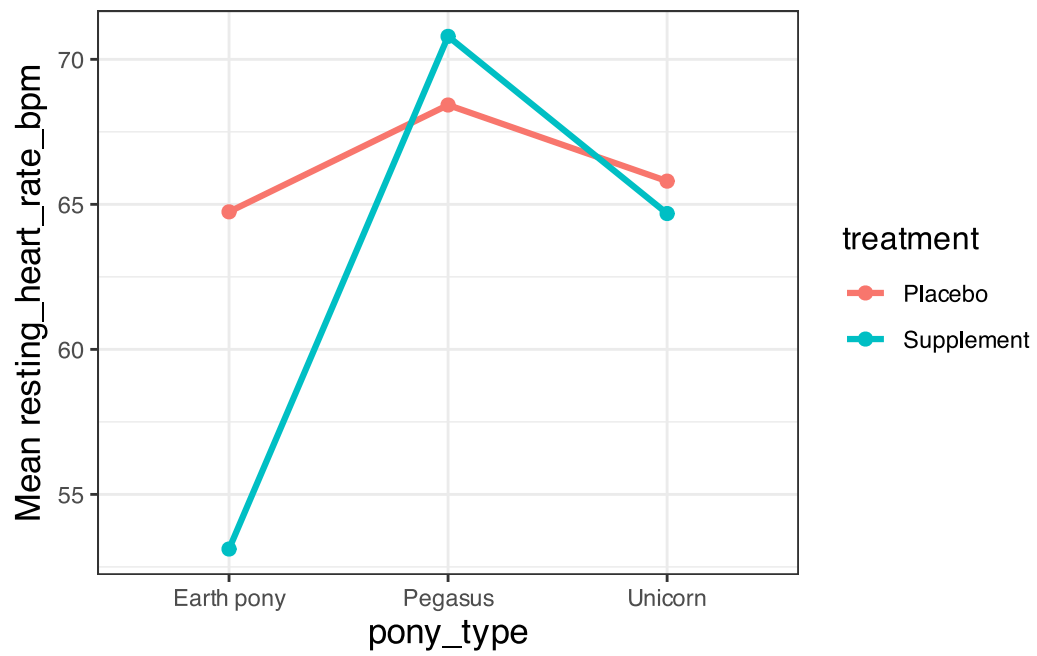
Main effect: pony_type

Comparison	Difference	p-value
Pegasus vs Earth pony	10.6792	< 0.001 *
Unicorn vs Earth pony	6.3125	< 0.001 *
Unicorn vs Pegasus	-4.3667	< 0.001 *

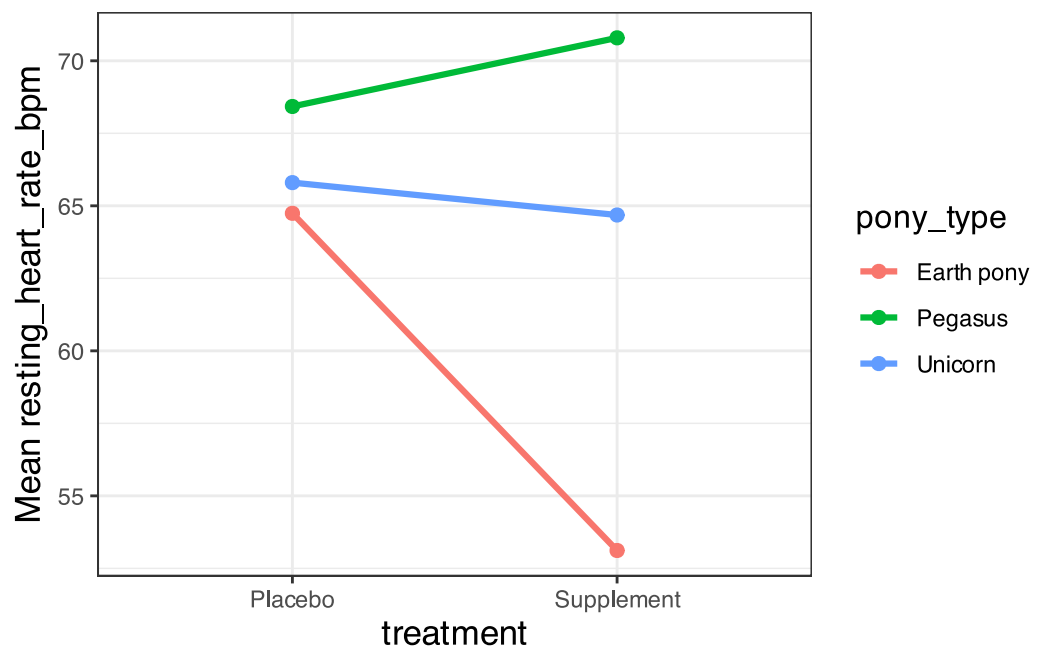
Interaction: treatment × pony_type		
Comparison	Difference	p-value
Supplement:Earth pony vs Placebo:Earth pony	-11.6250	< 0.001 *
Placebo:Pegasus vs Placebo:Earth pony	3.6833	0.167
Supplement:Pegasus vs Placebo:Earth pony	6.0500	0.002 *
Placebo:Unicorn vs Placebo:Earth pony	1.0583	0.982
Supplement:Unicorn vs Placebo:Earth pony	-0.0583	1.000
Placebo:Pegasus vs Supplement:Earth pony	15.3083	< 0.001 *
Supplement:Pegasus vs Supplement:Earth pony	17.6750	< 0.001 *
Placebo:Unicorn vs Supplement:Earth pony	12.6833	< 0.001 *
Supplement:Unicorn vs Supplement:Earth pony	11.5667	< 0.001 *
Supplement:Pegasus vs Placebo:Pegasus	2.3667	0.633
Placebo:Unicorn vs Placebo:Pegasus	-2.6250	0.524
Supplement:Unicorn vs Placebo:Pegasus	-3.7417	0.154
Placebo:Unicorn vs Supplement:Pegasus	-4.9917	0.020 *
Supplement:Unicorn vs Supplement:Pegasus	-6.1083	0.002 *
Supplement:Unicorn vs Placebo:Unicorn	-1.1167	0.977

* $p < \alpha = 0.050$

```
supplement_heart_rate |> profile_plot(outcome = resting_heart_rate_bpm,
                                       lines = treatment,
                                       x = pony_type)
```



```
supplement_heart_rate |> profile_plot(outcome = resting_heart_rate_bpm,
  lines = pony_type,
  x = treatment)
```



Example 2

```
exercise_recovery |> anovas(outcome = recovery_time_min,  
                             A = treatment,  
                             B = pony_type,  
                             interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	2	2909.4913	1454.7456	80.254	< 0.001
pony_type	2	23.1424	11.5712	0.638	0.530
Residuals	103	1867.0562	18.1268		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(2, 103) = 80.254$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):

H_0 : All means across levels of pony_type are equal

H_1 : At least one mean across levels of pony_type differs

Test statistic: $F(2, 103) = 0.638$

p-value: 0.530

Conclusion: Fail to reject the null hypothesis ($p = 0.530 \geq \alpha = 0.050$)

```
exercise_recovery |> posthoc(outcome = recovery_time_min,  
                             A = treatment,  
                             B = pony_type,  
                             interaction = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (recovery_time_min ~ treatment + pony_type)

Main effect: treatment

Comparison	Difference	p-value	
New treatment vs Standard care	-5.0806	< 0.001	*
Enhanced treatment vs Standard care	-12.6333	< 0.001	*
Enhanced treatment vs New treatment	-7.5528	< 0.001	*

Main effect: pony_type

Comparison	Difference	p-value
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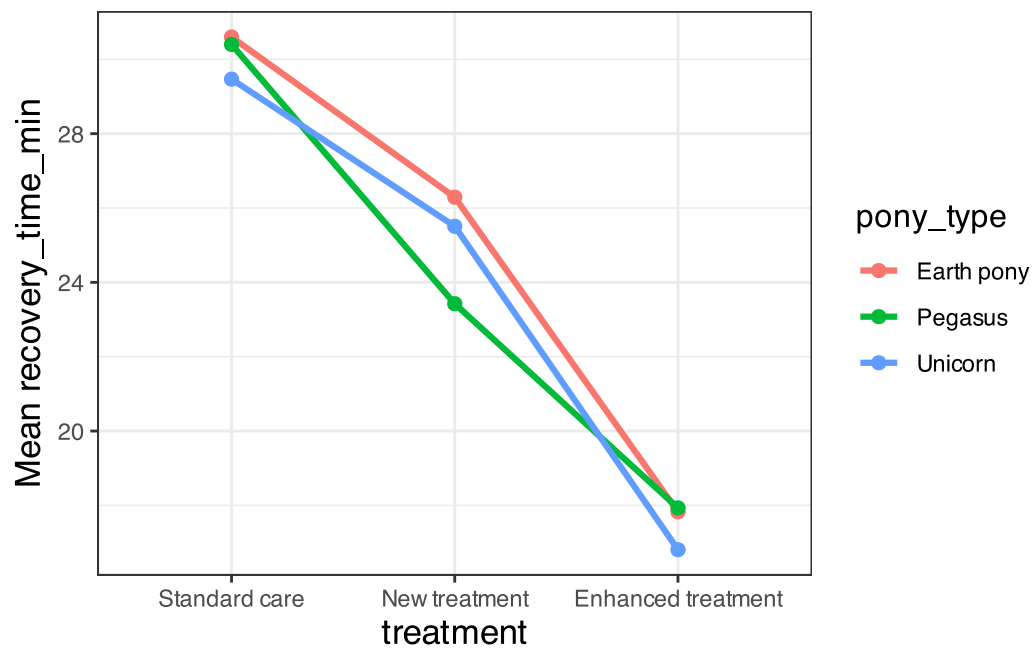
Pegasus vs Earth pony	-0.9861	0.589
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Unicorn vs Earth pony	-0.9778	0.595
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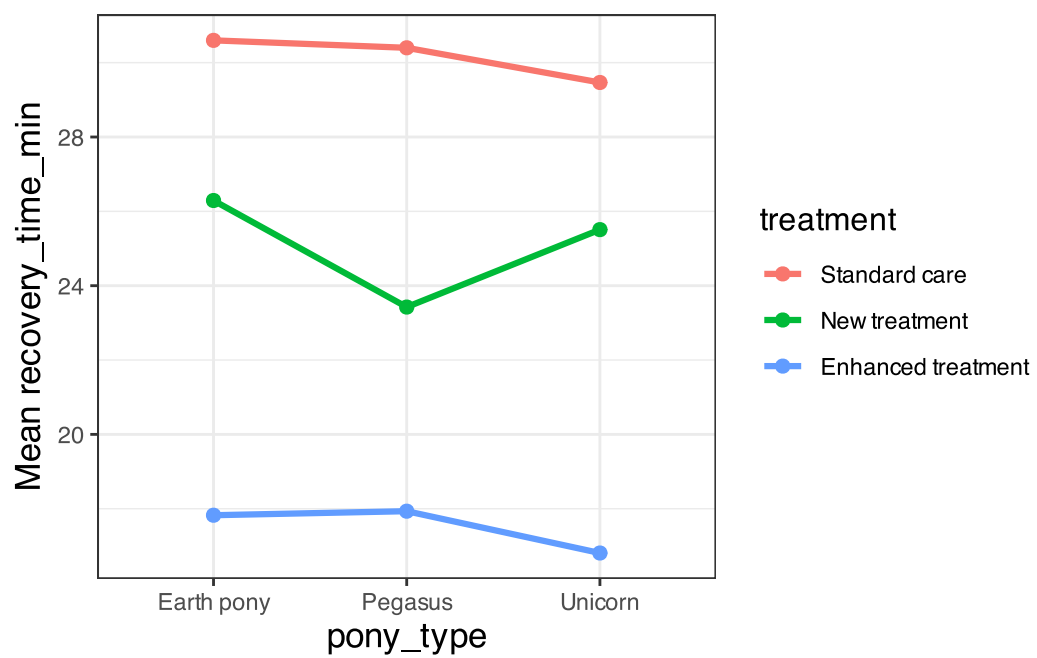
Unicorn vs Pegasus	0.0083	1.000
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* $p < \alpha = 0.050$

```
exercise_recovery |> profile_plot(outcome = recovery_time_min,  
                                xaxis = treatment,  
                                lines = pony_type)
```



```
exercise_recovery |> profile_plot(outcome = recovery_time_min,  
                                xaxis = pony_type,  
                                lines = treatment)
```



Let's Practice 1

Applejack and the Ponyville clinic are studying whether a high-fiber feed changes postmeal blood glucose. Within each pony type, adult ponies are randomly assigned to Standard feed or High-fiber feed for two weeks. After a standardized meal, glucose is measured once at the same pre-specified time for every pony.

```
glucose |> anovas(outcome = postmeal_glucose_mg_dl,  
                  A = treatment,  
                  B = pony_type)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	1	531.4410	531.4410	13.231	< 0.001
pony_type	2	2111.6527	1055.8263	26.286	< 0.001
treatment:pony_type	2	2635.6647	1317.8323	32.809	< 0.001
Residuals	84	3374.0227	40.1669		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(2, 84) = 32.809$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
glucose |> posthoc(outcome = postmeal_glucose_mg_dl,  
                  A = treatment,  
                  B = pony_type)
```

Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (postmeal_glucose_mg_dl ~ treatment * pony_type)

Main effect: treatment

Comparison	Difference	p-value
Standard feed vs High vs fiber feed	4.8600	< 0.001 *

Main effect: pony_type

Comparison	Difference	p-value
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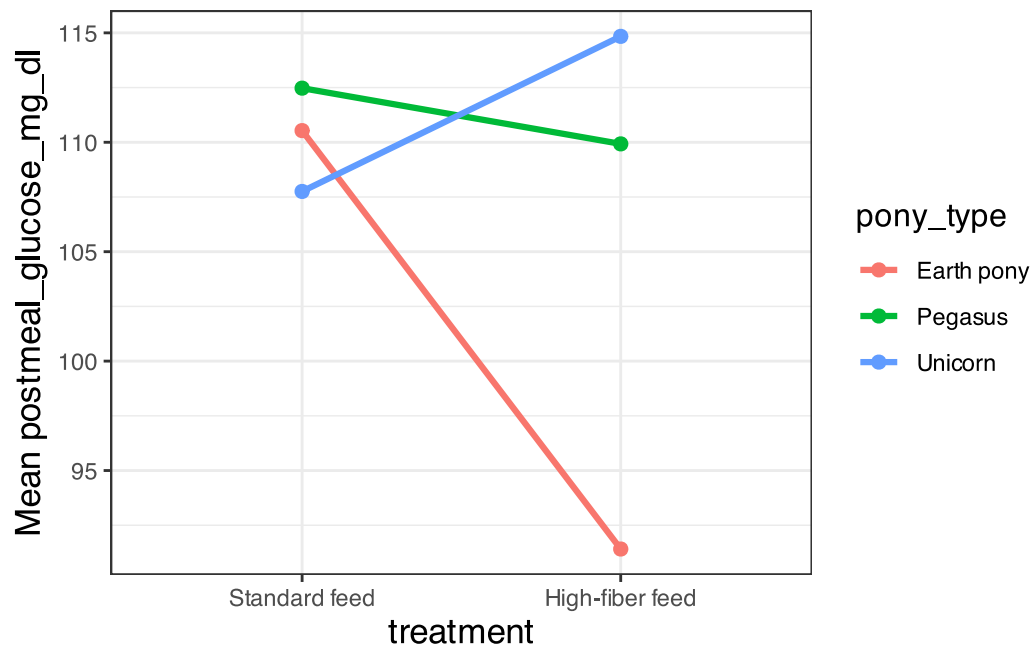
Pegasus vs Earth pony	10.2267	< 0.001	*
Unicorn vs Earth pony	10.3233	< 0.001	*
Unicorn vs Pegasus	0.0967	0.998	

Interaction: treatment × pony_type
Comparison

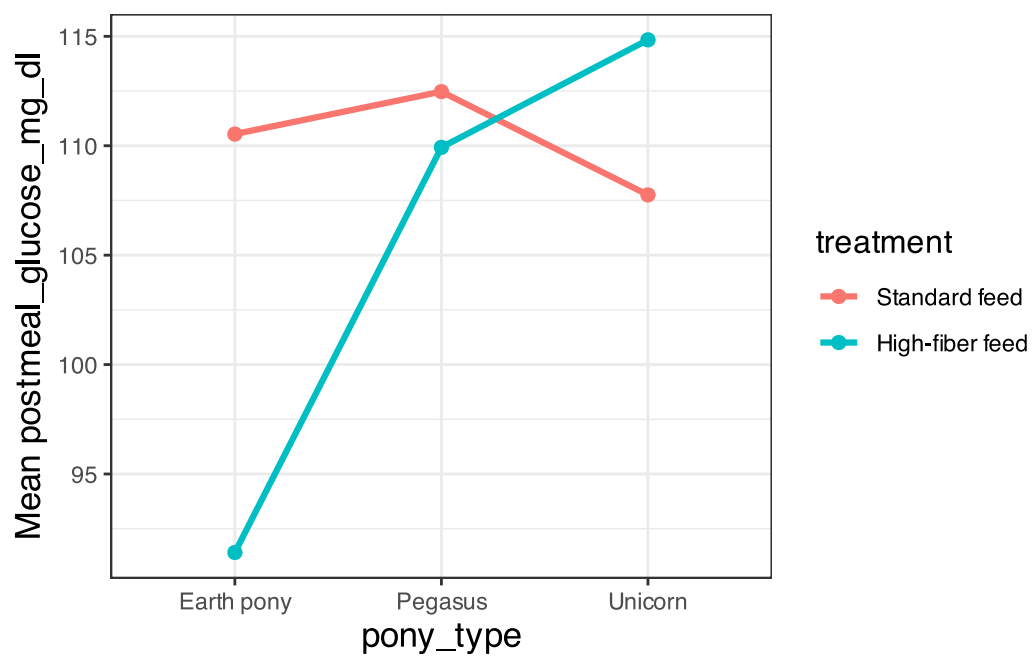
	Difference	p-value
Standard feed:Earth pony vs High vs fiber feed:Earth pony *	19.1200	< 0.001
High vs fiber feed:Pegasus vs High vs fiber feed:Earth pony *	18.5133	< 0.001
Standard feed:Pegasus vs High vs fiber feed:Earth pony *	21.0600	< 0.001
High vs fiber feed:Unicorn vs High vs fiber feed:Earth pony *	23.4267	< 0.001
Standard feed:Unicorn vs High vs fiber feed:Earth pony *	16.3400	< 0.001
High vs fiber feed:Pegasus vs Standard feed:Earth pony	-0.6067	1.000
Standard feed:Pegasus vs Standard feed:Earth pony	1.9400	0.959
High vs fiber feed:Unicorn vs Standard feed:Earth pony	4.3067	0.433
Standard feed:Unicorn vs Standard feed:Earth pony	-2.7800	0.835
Standard feed:Pegasus vs High vs fiber feed:Pegasus	2.5467	0.880
High vs fiber feed:Unicorn vs High vs fiber feed:Pegasus	4.9133	0.286
Standard feed:Unicorn vs High vs fiber feed:Pegasus	-2.1733	0.935
High vs fiber feed:Unicorn vs Standard feed:Pegasus	2.3667	0.909
Standard feed:Unicorn vs Standard feed:Pegasus	-4.7200	0.329
Standard feed:Unicorn vs High vs fiber feed:Unicorn *	-7.0867	0.034

* $p < \alpha = 0.050$

```
glucose |> profile_plot(outcome = postmeal_glucose_mg_dl,
                        xaxis = treatment,
                        lines = pony_type)
```



```
glucose |> profile_plot(outcome = postmeal_glucose_mg_dl,  
  axis = pony_type,  
  lines = treatment)
```



Let's Practice 2

Fluttershy is testing a fictional anti-inflammatory treatment in adult ponies with the same mild inflammatory condition. Within each pony type, ponies are randomly assigned to Placebo, Low dose, or High dose. A blood inflammation marker is measured once after seven days.

```
inflammation |> anovas(outcome = inflammation_marker_mg_l,  
                      A = treatment,  
                      B = pony_type)
```

Two-way ANOVA (with interaction)

Source	df	SS	MS	F	p-value
treatment	2	384.4924	192.2462	86.419	< 0.001
pony_type	2	366.1000	183.0500	82.286	< 0.001
treatment:pony_type	4	303.1356	75.7839	34.067	< 0.001
Residuals	126	280.2960	2.2246		

Hypotheses (treatment:pony_type):

H_0 : There is no interaction between treatment and pony_type

H_1 : There is an interaction between treatment and pony_type

Test statistic: $F(4, 126) = 34.067$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
inflammation |> posthoc(outcome = inflammation_marker_mg_l,  
                      A = treatment,  
                      B = pony_type)
```

Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (inflammation_marker_mg_l ~ treatment * pony_type)

Main effect: treatment

Comparison	Difference	p-value	
Low dose vs High dose	0.9378	0.010	*
Placebo vs High dose	3.9556	< 0.001	*
Placebo vs Low dose	3.0178	< 0.001	*

Main effect: pony_type

Comparison	Difference	p-value
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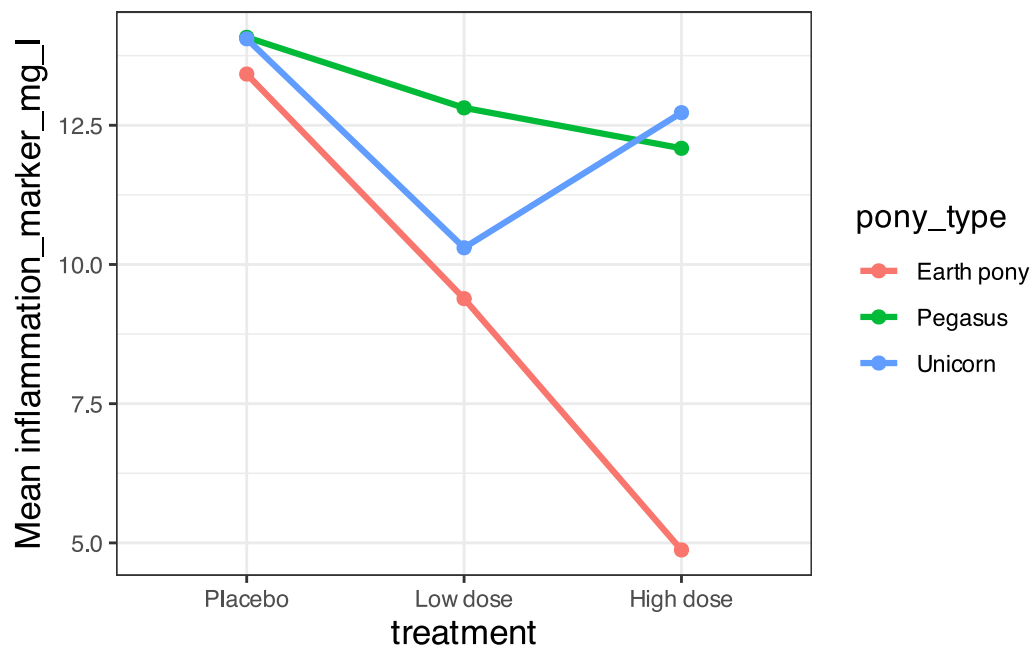
Pegasus vs Earth pony	3.7667	< 0.001	*
Unicorn vs Earth pony	3.1333	< 0.001	*
Unicorn vs Pegasus	-0.6333	0.113	

Interaction: treatment × pony_type
Comparison

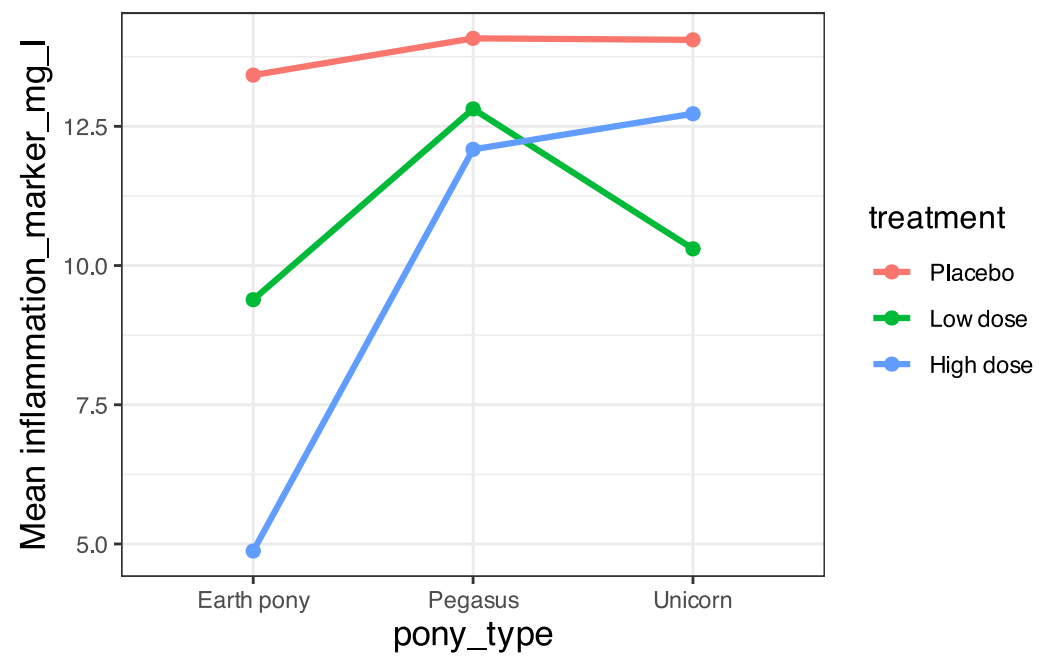
	Difference	p-value	
Low dose:Earth pony vs High dose:Earth pony	4.5133	< 0.001	*
Placebo:Earth pony vs High dose:Earth pony	8.5467	< 0.001	*
High dose:Pegasus vs High dose:Earth pony	7.2133	< 0.001	*
Low dose:Pegasus vs High dose:Earth pony	7.9400	< 0.001	*
Placebo:Pegasus vs High dose:Earth pony	9.2067	< 0.001	*
High dose:Unicorn vs High dose:Earth pony	7.8533	< 0.001	*
Low dose:Unicorn vs High dose:Earth pony	5.4267	< 0.001	*
Placebo:Unicorn vs High dose:Earth pony	9.1800	< 0.001	*
Placebo:Earth pony vs Low dose:Earth pony	4.0333	< 0.001	*
High dose:Pegasus vs Low dose:Earth pony	2.7000	< 0.001	*
Low dose:Pegasus vs Low dose:Earth pony	3.4267	< 0.001	*
Placebo:Pegasus vs Low dose:Earth pony	4.6933	< 0.001	*
High dose:Unicorn vs Low dose:Earth pony	3.3400	< 0.001	*
Low dose:Unicorn vs Low dose:Earth pony	0.9133	0.759	
Placebo:Unicorn vs Low dose:Earth pony	4.6667	< 0.001	*
High dose:Pegasus vs Placebo:Earth pony	-1.3333	0.268	
Low dose:Pegasus vs Placebo:Earth pony	-0.6067	0.971	
Placebo:Pegasus vs Placebo:Earth pony	0.6600	0.953	
High dose:Unicorn vs Placebo:Earth pony	-0.6933	0.937	
Low dose:Unicorn vs Placebo:Earth pony	-3.1200	< 0.001	*
Placebo:Unicorn vs Placebo:Earth pony	0.6333	0.963	
Low dose:Pegasus vs High dose:Pegasus	0.7267	0.919	
Placebo:Pegasus vs High dose:Pegasus	1.9933	0.011	*
High dose:Unicorn vs High dose:Pegasus	0.6400	0.960	
Low dose:Unicorn vs High dose:Pegasus	-1.7867	0.035	*
Placebo:Unicorn vs High dose:Pegasus	1.9667	0.013	*
Placebo:Pegasus vs Low dose:Pegasus	1.2667	0.335	
High dose:Unicorn vs Low dose:Pegasus	-0.0867	1.000	
Low dose:Unicorn vs Low dose:Pegasus	-2.5133	< 0.001	*
Placebo:Unicorn vs Low dose:Pegasus	1.2400	0.364	
High dose:Unicorn vs Placebo:Pegasus	-1.3533	0.249	
Low dose:Unicorn vs Placebo:Pegasus	-3.7800	< 0.001	*
Placebo:Unicorn vs Placebo:Pegasus	-0.0267	1.000	
Low dose:Unicorn vs High dose:Unicorn	-2.4267	< 0.001	*
Placebo:Unicorn vs High dose:Unicorn	1.3267	0.274	
Placebo:Unicorn vs Low dose:Unicorn	3.7533	< 0.001	*

* $p < \alpha = 0.050$

```
inflammation |> profile_plot(outcome = inflammation_marker_mg_l,
                             xaxis = treatment,
                             lines = pony_type)
```



```
inflammation |> profile_plot(outcome = inflammation_marker_mg_l,
                             lines = treatment,
                             xaxis = pony_type)
```



Let's Practice 3

Nurse Redheart compares dressings in adult ponies receiving care for comparable minor wounds. Within each pony type, ponies are randomly assigned to Standard dressing, Hydrogel, or Enhanced hydrogel. The clinic records the time from first dressing to complete wound closure using a common clinical criterion. Each pony contributes one wound.

```
wound_healing |> anovas(outcome = wound_closure_days,  
                        A = treatment,  
                        B = pony_type,  
                        interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	2	1306.4801	653.2401	244.050	< 0.001
pony_type	2	944.2704	472.1352	176.390	< 0.001
Residuals	130	347.9661	2.6767		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(2, 130) = 244.05$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):

H_0 : All means across levels of pony_type are equal

H_1 : At least one mean across levels of pony_type differs

Test statistic: $F(2, 130) = 176.39$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

```
wound_healing |> posthoc(outcome = wound_closure_days,  
                        A = treatment,  
                        B = pony_type,  
                        interaction = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA ($\text{wound_closure_days} \sim \text{treatment} + \text{pony_type}$)

Main effect: treatment

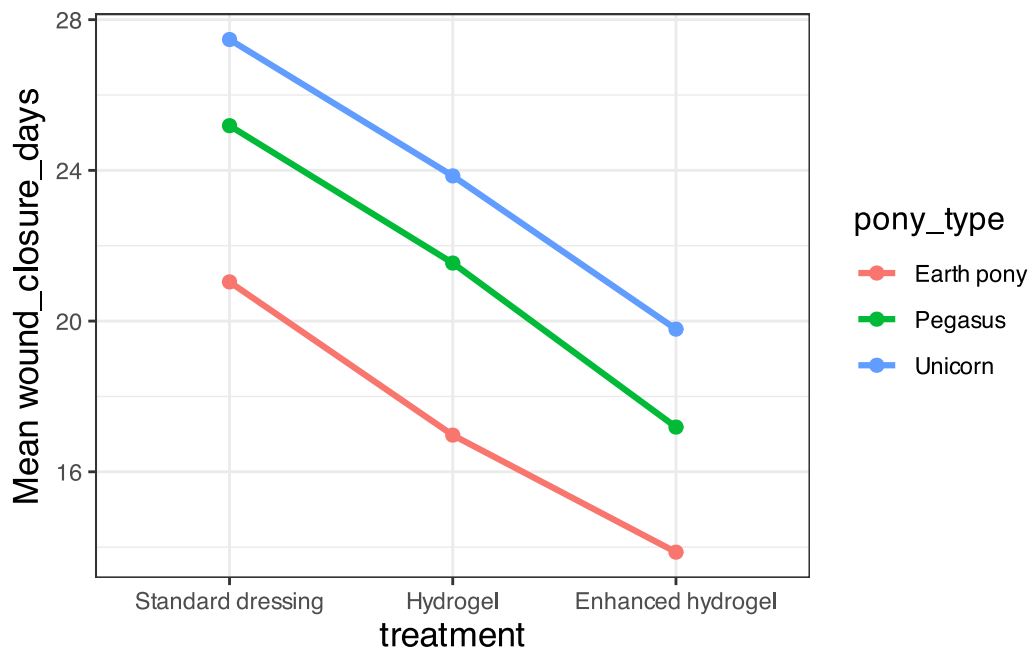
Comparison	Difference	p-value
Hydrogel vs Enhanced hydrogel	3.8422	< 0.001 *
Standard dressing vs Enhanced hydrogel	7.6200	< 0.001 *
Standard dressing vs Hydrogel	3.7778	< 0.001 *

Main effect: pony_type

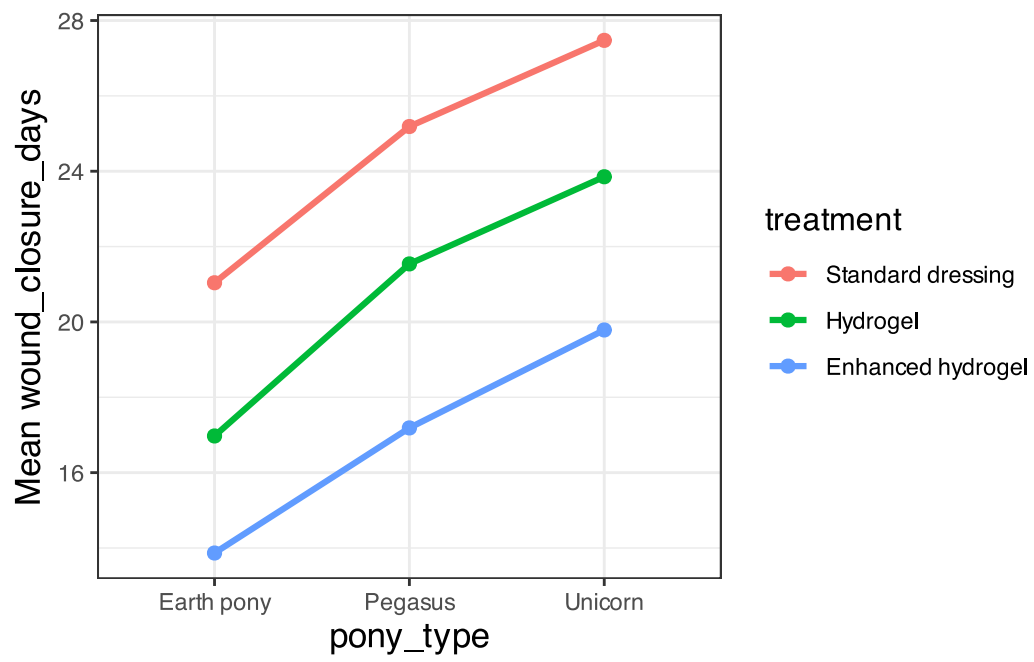
Comparison	Difference	p-value
Pegasus vs Earth pony	4.0111	< 0.001 *
Unicorn vs Earth pony	6.4111	< 0.001 *
Unicorn vs Pegasus	2.4000	< 0.001 *

* $p < \alpha = 0.050$

```
wound_healing |> profile_plot(outcome = wound_closure_days,
                             xaxis = treatment,
                             lines = pony_type)
```



```
wound_healing |> profile_plot(outcome = wound_closure_days,
                             lines = treatment,
                             xaxis = pony_type)
```



Let's Practice 4

Rainbow Dash helps the Ponyville clinic compare two fluid treatments after a standardized exercise session. Within each pony type, adult ponies are randomly assigned to Standard fluids or Electrolyte fluids. Time to reach a prespecified hydration target is recorded once for each pony.

```
rehydration |> anovas(outcome = rehydration_time_min,  
                     A = treatment,  
                     B = pony_type,  
                     interaction = FALSE)
```

Two-way ANOVA (no interaction)

Source	df	SS	MS	F	p-value
treatment	1	2301.2890	2301.2890	264.232	< 0.001
pony_type	2	36.5502	18.2751	2.098	0.129
Residuals	86	749.0040	8.7093		

Hypotheses (treatment):

H_0 : All means across levels of treatment are equal

H_1 : At least one mean across levels of treatment differs

Test statistic: $F(1, 86) = 264.232$

p-value: < 0.001

Conclusion: Reject the null hypothesis ($p = < 0.001 < \alpha = 0.050$)

Hypotheses (pony_type):

H_0 : All means across levels of pony_type are equal

H_1 : At least one mean across levels of pony_type differs

Test statistic: $F(2, 86) = 2.098$

p-value: 0.129

Conclusion: Fail to reject the null hypothesis ($p = 0.129 \geq \alpha = 0.050$)

```
rehydration |> posthoc(outcome = rehydration_time_min,  
                     A = treatment,  
                     B = pony_type,  
                     interaction = FALSE)
```

Post-hoc analysis: Tukey's HSD

Following: two-way ANOVA (rehydration_time_min ~ treatment + pony_type)

Main effect: treatment

Comparison	Difference	p-value
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Standard fluids vs Electrolyte fluids	10.1133	< 0.001	*
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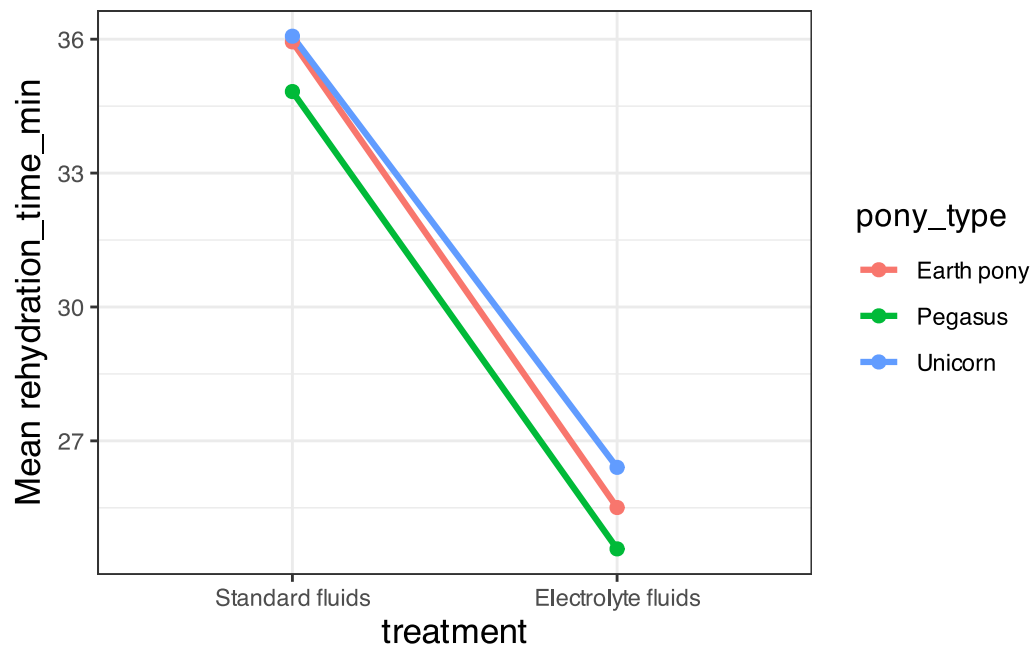
Main effect: pony_type

Comparison	Difference	p-value
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Pegasus vs Earth pony	-1.0200	0.378
Unicorn vs Earth pony	0.5133	0.779
Unicorn vs Pegasus	1.5333	0.115

* $p < \alpha = 0.050$

```
rehydration |> profile_plot(outcome = rehydration_time_min,
                           xaxis = treatment,
                           lines = pony_type)
```



```
rehydration |> profile_plot(outcome = rehydration_time_min,
                           lines = treatment,
                           xaxis = pony_type)
```

